

ARTI403: Image Processing

Lab 2: Digital Image Fundamentals

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Objective

To explain how digital images are represented and manipulated in a computer.

Tools

Python 3, Jupyter, OpenCV, NumPy, Matplotlib.

Procedural Steps and Tasks

I used two grayscale images: camera.png and coins.png. I resized coins.png to 512×512 so both images have the same size.



Figure 1: The two images used

Task 1: Sampling and Quantization

The manual uses sampling factor 14 and 9 quantization levels. I changed them to 7 and 16 and compared the results.

```
sampld    = sample_image(img1, 7)
quantized = quantize_image(img1, 16)
```

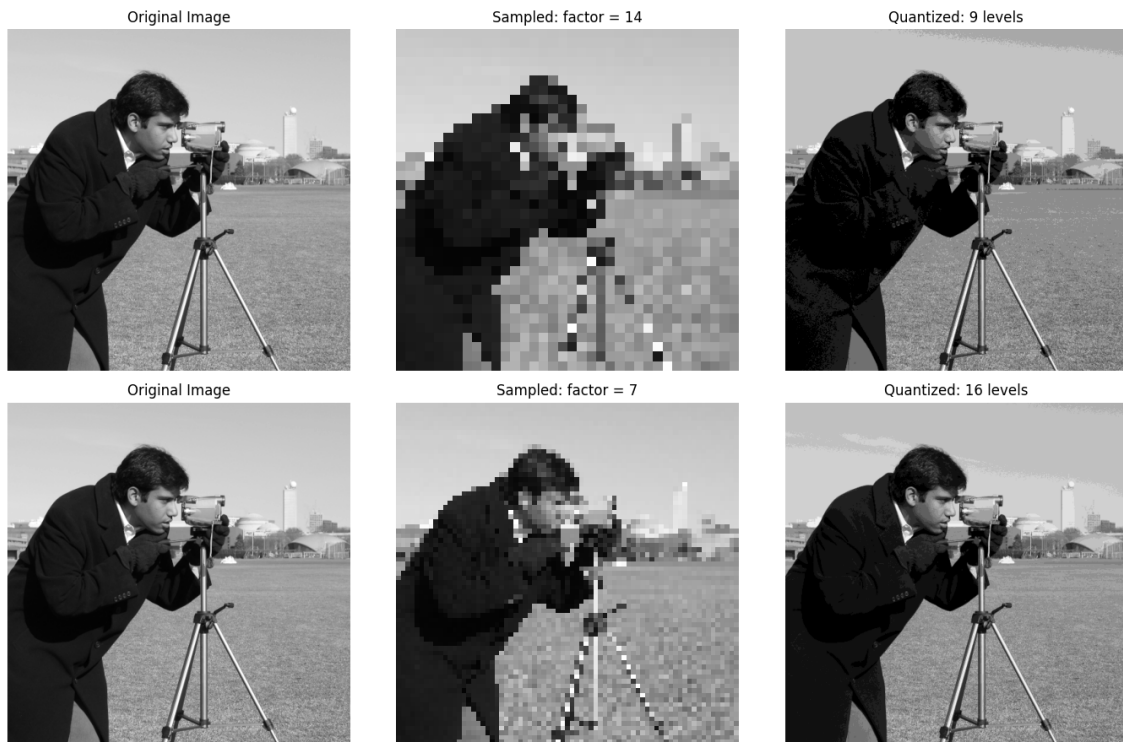


Figure 2: Top row: factor 14 and 9 levels. Bottom row: factor 7 and 16 levels

- A bigger sampling factor makes the image smaller and removes more details.
- More quantization levels keep more gray shades, so the image looks smoother.

Task 2: Operations on Two Images

I converted both images to arrays and did the following operations.

Arithmetic operations

```
addition      = im1arr + im2arr          # example from the manual
subtraction   = cv2.subtract(im1arr, im2arr) # subtract two images
add_175       = cv2.add(im1arr, np.full_like(im1arr, 175)) # add constant 175
```

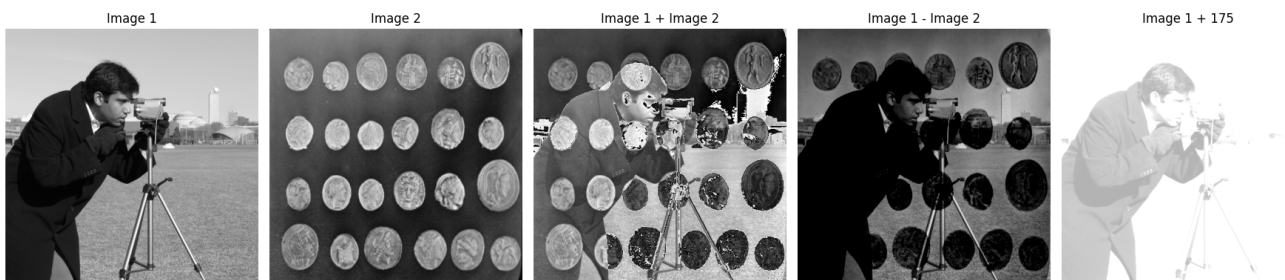


Figure 3: Addition, subtraction and adding 175

Adding 175 makes the image very bright because the values are limited to a maximum of 255.

Set and logical operations

```
union          = np.bitwise_or(A, B)          # example from the manual
set_diff       = np.bitwise_and(A, np.bitwise_not(B)) # set difference
sym_diff       = np.bitwise_xor(A, B)          # symmetric difference
intersection    = np.bitwise_and(A, B)          # intersection
```

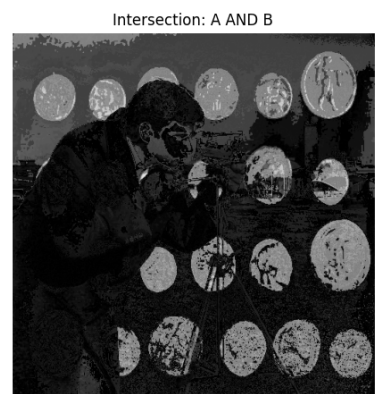
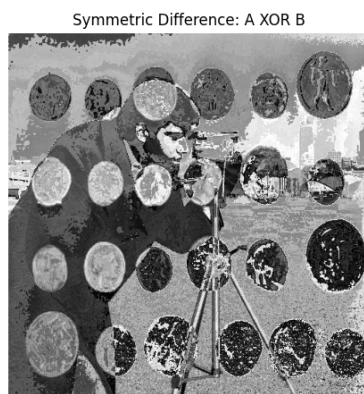
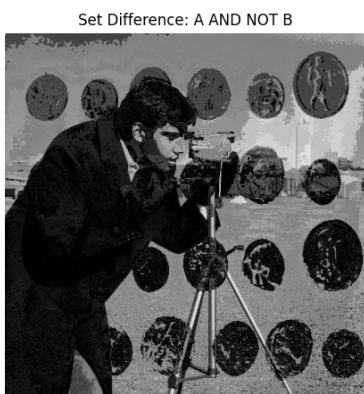
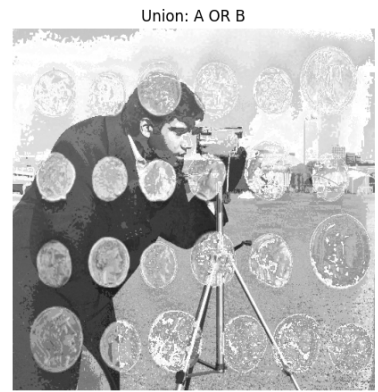
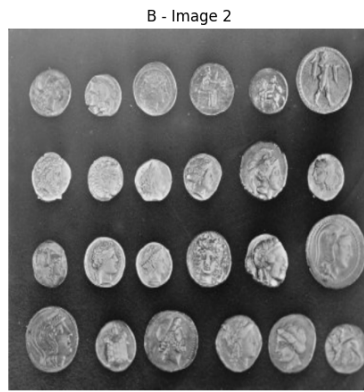


Figure 4: Union, set difference, symmetric difference and intersection